A CENTRALIZED SYSTEM FOR REAL-TIME FARM ANIMAL DISEASE REPORTING

INDIVIDUAL THESIS

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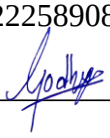
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ABSTRACT

Timely spatial interpretation of livestock disease reports remains difficult where outbreak records are delayed, geographically imprecise and distributed across independent systems. This individual research develops a geospatial clustering and trajectory-mapping component within the Animal Disease Reporting System (ADRS) for Sri Lankan veterinary surveillance. The component combines provenance-controlled historical outbreak processing, outbreak-triggered spatio-temporal analysis, a geodesic spatial ranking model, interpretable direction and reach descriptors, and authenticated veterinary map interfaces. Historical lumpy skin disease records from EMPRES-i and WAHIS were consolidated from 3,089 raw records into 1,480 model-eligible candidates; foot-and-mouth disease was treated through a separate origin-level analysis path. A frozen exponential distance-kernel model ranked 5 km spatial cells for follow-on events in an evaluated D1–D7 retrospective window. The recorded held-out evaluation obtained a mean area-weighted target percentile of 73.85 (95% interval 70.51–77.11), conditional on its 25 km evaluation scope. A descriptive geometric direction field passed a structural audit of 560,853 cells without invariant failures, while a historical apparent spread-front estimate of 3.95 km/day provided nominal reach context. A React/MapLibre interface presents national confirmed-case views, district-focused My Area mapping, a fourteen-day illustrative risk-outlook timeline, source inspection and Analysis & Trends. Scientific model results are distinguished from presentation-layer projections and clinical recommendations. The work demonstrates a traceable architecture for converting historical and veterinarian-confirmed outbreak observations into interpretable spatial decision support, while identifying the additional local data and prospective validation required for operational predictive use.

Keywords: Livestock Disease Surveillance, Spatial Epidemiology, Geospatial Risk Ranking, Trajectory Mapping, Veterinary Decision Support

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Finally, I acknowledge the work of the World Organisation for Animal Health, the Food and Agriculture Organization, open geospatial data providers and the scientific software communities whose public documentation, tools and research materials contributed to this study. Responsibilities for the individual analyses, interpretations and remaining limitations rest with the author.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ADRS	Animal Disease Reporting System
API	Application Programming Interface
CI	Confidence Interval
C0	Frozen distance-kernel spatial ranking model
DAPH	Department of Animal Production and Health
DQS	Data Quality Score
EMPRES-i	Emergency Prevention System – global animal disease information
FMD	Foot-and-Mouth Disease
GIS	Geographic Information System
JWT	JSON Web Token
KDE	Kernel Density Estimation
LSD	Lumpy Skin Disease
ML	Machine Learning
PISTES	Project framework label used in the technical repository
S0	Apparent historical spread-front rate estimator
SSE	Server-Sent Events
ST-DBSCAN	Spatial-Temporal Density-Based Spatial Clustering of Applications with Noise
UTM	Universal Transverse Mercator
WAHIS	World Animal Health Information System
WOAH	World Organisation for Animal Health
WS	WebSocket

1 INTRODUCTION

1.1 Overall project context

Livestock farming contributes directly to household livelihoods, food supply chains and agricultural resilience in Sri Lanka. Transboundary and endemic animal diseases can interrupt milk production, trade and animal movement, making early reporting and rapid geographical interpretation important parts of veterinary practice. Foot-and-mouth disease (FMD) is a highly contagious disease of cloven-hoofed livestock; lumpy skin disease (LSD) affects cattle and buffalo and is commonly associated with arthropod vectors [1], [2]. Consequently, the temporal position of an event, its geographical setting and the evidence supporting a veterinary confirmation matter as much as the disease name recorded in a database.

The group research project R26-SE-013, A Centralized System for Real-Time Farm Animal Disease Reporting, proposes ADRS as a shared environment for reporting, veterinary review and decision support. The project proposal identifies the consequences of fragmented reporting, delayed aggregation and the absence of an integrated visual overview. The full platform incorporates components developed by different team members. This individual thesis is confined to the geospatial research, engineering and interface that converts authorized case information and historical outbreak records into spatial context for veterinary officers.

1.2 Individual component context

The individual component, Real-Time Geospatial Clustering & Trajectory Mapping, is organized around a practitioner-facing workflow. After authentication, a veterinarian can open a national outbreak map, inspect geographically positioned and vet-confirmed clinical cases, select relevant historical outbreak context and focus on an assigned district. The My Area interface provides local spatial context, a time-dependent outlook presentation, source information, risk-level graphics and links to operational farm information. Analysis & Trends provides a complementary retrospective view. Figure 1.1 summarizes the intended journey from verified reporting to geographical decision support.

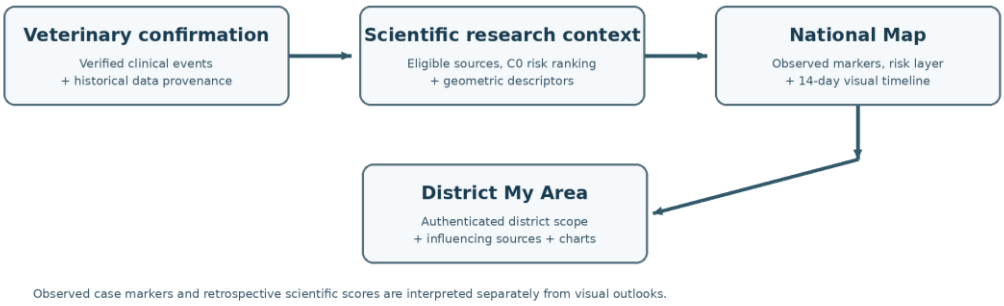


Figure 1.1 Practitioner workflow from validated source evidence to national and district-level geospatial views.

Source: Author's schematic synthesis of the recorded application workflow and component design.

The word "real-time" in the project title refers to the intention to reduce delay in surfacing newly verified clinical records. It must not be confused with independently validated, continuous forecasting of transmission. The implementation separates live operational markers from a historical research-analysis domain; the latter computes relative spatial ranks for historical forecast origins. That separation is central to the scientific interpretation of all map layers and to the responsible presentation of a fourteen-day outlook.

1.3 Problem statement

A list of confirmed cases alone does not tell a veterinarian how geographically concentrated the observed events are, which historical source records explain a displayed pattern, how the relative spatial attention areas compare, or which farms may need inspection. This problem is compounded by inconsistent historical date fields, uncertain geographic precision and duplication between international surveillance sources. Directly fitting a predictive model to all records without controlling those issues risks target leakage, artificially enlarged outbreak counts and unwarranted confidence in reported locations.

The individual technical problem is therefore to create a traceable pathway from vetted case observations and imperfect historical outbreak records to intelligible geospatial displays, while preserving the distinction among observed cases, historical spatial rankings, geometric direction descriptors and illustrative forward visualisations. In an operational setting, a district veterinarian needs both the national

picture and a geographically restricted view, without treating every map colour as a measured probability of future infection.

1.4 Research gap

International animal-health platforms provide validated data and map-based access at broad geographical scales [3], [4]. Research on density-based spatio-temporal clustering and kernel modelling gives suitable techniques for analyzing spatial and temporal patterns [5], [6]. However, these strands alone do not provide the integrated practitioner workflow required by the ADRS proposal: provenance-aware historical processing, role-aware live case overlay, explicit separation of source and target time, direction/reach context and veterinary district interfaces in one connected product. The project's individual research gap is an integration-and-interpretability gap, not a claim that any of these underlying mathematical techniques is newly invented.

A further gap arises when environmental variables are attractive in theory but unavailable consistently at the time and location of each outbreak. In such circumstances, a transparent spatial baseline with properly described scope is more defensible than a visually sophisticated hybrid model whose required features cannot be reproduced. This thesis accordingly examines the historical evidence supporting the selected distance-kernel baseline and describes where environmental integrations remain an evidence-dependent extension.

1.5 Aim and objectives

Aim. To design, implement and evaluate an interpretable geospatial intelligence component for Sri Lankan veterinary surveillance that connects historical livestock outbreak analysis with vet-confirmed operational cases and district-focused visual decision support.

ID	Specific objective
O1	Build a provenance-controlled historical outbreak dataset with disease, date, coordinate, quality and duplicate-handling rules.
O2	Implement spatio-temporal context selection and a reproducible spatial scoring framework for historical forecast origins.

ID	Specific objective
O3	Derive interpretable relative spatial ranks, geometric direction context and a bounded historical reach descriptor.
O4	Provide national and veterinarian-area interfaces with confirmed case inspection, timeline controls and source/farm context.
O5	Evaluate scientific outputs and software contracts separately and document practical limitations for responsible use.

Table 1.1 Individual research objectives.

1.6 Research questions or hypotheses

RQ1 asks how historical animal-disease reports from heterogeneous systems can be consolidated into a reproducible, leakage-aware set of spatial research events. RQ2 asks whether an interpretable distance-based model can prioritize historical D1–D7 follow-on event locations within a declared evaluation domain. RQ3 asks how a geometric direction tendency and a historical spread-rate descriptor can be communicated without misrepresenting them as validated transmission vectors. RQ4 asks how these results and operational confirmations should be represented in a national and district-specific veterinarian interface. These questions are evaluated using code artefacts, recorded experiments and observed interface behaviour rather than unsupported user-impact assertions.

1.7 Scope, limitations and assumptions

The component's research domain covers LSD and FMD, but the two disease pathways have different outputs. The documented LSD analysis supports a spatial grid, relative C0 score, derived direction and nominal reach. The FMD pathway uses a separate origin-level scalar-risk candidate; the code audit did not establish a corresponding validated FMD direction-field or spatial trajectory. Research development relied heavily on non-Sri-Lankan historical records, with Sri Lanka considered through a deliberately small descriptive transfer case. Operational map access is situated within the Sri Lankan veterinary portal and may consume clinical cases only after the host workflow marks them appropriately verified.

The recorded user interface includes a fourteen-day visual outlook, particularly for My Area and a Matara demonstration. Its coloured sequence and projected corridors

are presentation-layer constructs; the recorded scientific target evaluation instead pools D1–D7 outcomes. The thesis treats that fourteen-day display as an interaction and communication design and does not interpret it as fourteen independently validated disease incidence predictions. District assignment, available farm records and authority-approved response protocols are assumed inputs to deployment.

1.8 Individual contribution

The geospatial component's individual contribution includes historical-data adapters and canonicalization, geodesic domain/grid construction, source eligibility and research splits, a frozen distance-kernel ranking model, structural direction calculations, retrospective apparent-rate estimation, analytical endpoints and snapshot transfer, plus national mapping and district interfaces. The host system's authentication, clinical diagnosis, farm-management records and other prediction modules are shared team capabilities and are described here only at their integration boundaries.

1.9 Thesis structure

Chapter 2 examines related surveillance and spatial-analysis approaches. Chapter 3 specifies data provenance, modeling and validation methodology. Chapter 4 presents product and component design; Chapter 5 addresses technical implementation and integration tests. Chapter 6 reports recorded scientific findings and application evidence. Chapter 7 interprets the results and limitations, Chapter 8 considers adoption and sustainability, and Chapter 9 concludes the individual work and identifies future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review draws on established spatial-statistical literature, peer-reviewed methods, official animal-health data portals and documented geospatial software. The selection emphasizes five questions: the reliability of event source data, methods of detecting geographical concentration, construction of interpretable spatial surfaces, time-aware evaluation and communication of uncertainty to operational practitioners. Claims about the local software implementation are grounded separately in the project repository and its technical evidence audit. The reviewed sources provide theoretical and system context; they do not replace experimental validation of the ADRS implementation.

2.2 Domain and theoretical foundations

2.2.1 Spatial epidemiology and outbreak surveillance

Spatial epidemiology considers the geographic distribution of events in relation to population, environment and time. A recorded livestock outbreak combines a disease label, location, time and possibly a count of affected animals. Interpretation requires care because an incident can be approximately geocoded, reported after onset or repeated through more than one surveillance channel. The WOAH WAHIS platform distributes officially reported animal-health events and offers mapping and extraction capabilities; FAO's EMPRES-i ecosystem is another important disease-information source [3], [4]. As a result, access to data does not eliminate the need to document provenance and reconcile records before model fitting.

2.2.2 Spatio-temporal clustering

Density-based clustering identifies groups of nearby events and explicitly retains isolated points as noise. Birant and Kut extended such ideas into ST-DBSCAN, incorporating spatial and temporal criteria into neighborhood construction [5]. For veterinary events, using only a spatial radius can merge outbreaks separated by many months, while using only a temporal window can connect geographically unrelated reports. A paired criterion is more meaningful. Nevertheless, the resulting clusters are sensitive to radius and minimum-support choices, duplicate locations and the

limited spatial precision of reports. A cluster is evidence of concentration under defined settings, not a demonstrated chain of pathogen transmission.

2.2.3 Kernel methods and ranked spatial surfaces

Kernel estimation provides a mechanism to spread the contribution of each event over neighbouring geographical locations rather than displaying isolated points alone. Silverman's treatment establishes the general foundation of kernel smoothers and their scale dependence [6]. The individual project developed both Gaussian and exponential computational primitives but selected an exponential distance-only reference model for the evaluated LSD ranking. The purpose of this model is ordering cells by relative contextual score. It is not a calibrated probability surface and should not be labelled as the percentage chance that a particular farm will become infected.

2.2.4 Geospatial direction and environmental drivers

Directional fields are visually effective but scientifically ambiguous unless their construction is declared. The gradient of a spatial score describes the direction of greatest score increase; its negative points toward score decrease. Such a vector can provide coherent geometry for arrows even when neither animal movement nor environmental transport is measured. In contrast, biologically interpreted disease movement would require relevant wind, vector, livestock movement and outbreak observations, as well as independent target bearings for evaluation. WOAH's disease descriptions make clear that the physical pathways for FMD and LSD are not interchangeable [1], [2]. Therefore a single generic arrow should not imply equivalent propagation mechanisms for both diseases.

2.2.5 Temporal validation and uncertainty

Historical modelling should ensure that information used to construct a score would have been available at the simulated origin time. A random split of event rows can leak later observations from the same outbreak chain into model inputs. The research uses calendar-based development roles, an embargo around validation windows and explicit historical replay to reduce that risk. Confidence intervals are useful when paired with a clear sampling unit: a bootstrap over target events describes variation in a historical estimator under stated assumptions, whereas a measured prospective

prediction interval would require separate validation. Efron and Tibshirani provide a methodological basis for bootstrap inference [7].

2.3 Related research

The literature suggests a progression from event-mapping platforms to spatial pattern analysis and predictive surfaces. Administrative dashboards maximize regional visibility and data exchange. Cluster methods add interpretable event groups but may not yield a district-friendly forward-looking display. Kernel methods generate continuous surfaces but require well-defined scales and calibration before probability language is justified. Machine-learning approaches can combine many predictors but may become difficult to reproduce where environmental features are missing or inconsistently timed. The present study addresses those constraints by adopting a comparatively simple, auditable baseline and then building layered practitioner-oriented visualization on top of it.

Approach	Primary contribution	Limitation in this study's setting
WAHIS / global reporting	Officially reported events and global visualization	Reporting-time differences and variable geographical precision
ST-DBSCAN	Spatio-temporal event grouping with explicit noise	Sensitive to clustering parameters; not automatically transmission
Kernel spatial ranking	Interpretable distance-sensitive spatial field	Relative ordering is not calibrated infection probability
Environmental / ML prediction	Potential to incorporate exposure factors	Coverage and temporal-availability requirements
ADRS individual component	Historical analysis linked to live vet context and area interface	Prospective outcomes and local field evaluation remain essential

Table 2.1 Comparison of approaches relevant to the individual research.

2.4 Existing systems, products and techniques

WAHIS offers globally recognized official event reporting and geographic dashboards [3]. FAO's national and global EMPRES-i+ offerings distinguish public event intelligence from operational data governed by national veterinary services [4]. In the software environment, MapLibre GL JS provides a browser-based interactive geographical renderer, supporting map movement, symbols and layered spatial datasets [8]. None of these products should be described as having the same specific

authentication, local farm context and presentation behaviour as the ADRS demonstration without a formal comparison study.

The project's product design borrows the practical interaction pattern of layered GIS dashboards: an overview for coverage, drill-down for the local district, a timeline for interpreting changes and information panels for source inspection. This structure is useful because a veterinarian's decision task is not equivalent to reading a model score table. Colour, arrows and the distinction between observed and illustrative elements need consistent labelling so visual prominence does not imply additional evidential certainty.

2.5 Gap synthesis and conceptual framework

The research identifies three complementary requirements. First, scientific defensibility demands reproducible data origin, stable source eligibility and temporally separated evaluation. Second, interpretable spatial analytics demand an explicit mathematical meaning for every map score, vector and reach descriptor. Third, operational usability demands that the authorized veterinarian can connect national observations to an assigned geographic area, inspected sources and relevant animal holdings. ADRS's geospatial component contributes an implementation that connects these requirements while recording which layers are retrospective research outputs and which are interface illustrations.

The resulting conceptual framework is: raw or verified event → identity and provenance checks → eligible historical context or operational view → distance-based spatial ranking and geometric descriptors → domain-specific API representation → national or area-level map → veterinarian review and any authority-approved field action. The framework deliberately ends with professional review rather than autonomous control of farms or mandatory interventions.

3 RESEARCH METHODOLOGY

3.1 Research design

The study follows an artefact-oriented design science approach with repeated cycles of requirements analysis, data quality examination, model development, technical verification and interface refinement. The research artefact is not merely a map; it includes a canonical event definition, leakage guards, selected spatial-scoring rule, direction-and-rate derivations and application interfaces that expose the outcome. Evaluation is divided into scientific ranking evidence, mathematical/structural correctness and application contract/interaction evidence. This separation prevents a successful front-end demonstration from being counted as statistical validation of a future outbreak.

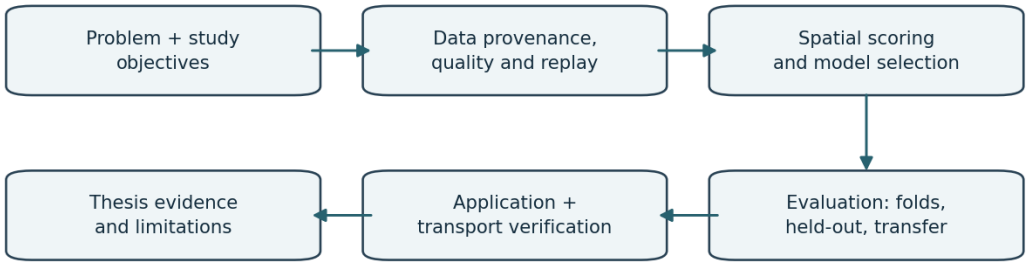


Figure 3.1 Research lifecycle: provenance and historical evaluation precede application communication.

Source: Author's methodology synthesis; stages follow the audited geospatial component.

A retrospective observational study design was selected because sufficient prospective geo-referenced field outcomes and operational availability timestamps were not present in the supplied evidence. It supports a reproducible examination of past source-target relationships under declared time rules. The user interface also functions as a design-and-demonstration artefact: the veterinarian can inspect confirmed case markers, a district outlook, source details and visual guidance. Those interface observations establish what was presented to a user, not whether future disease spread occurred as pictured.

3.2 Data, participants or study materials

3.2.1 Historical surveillance sources

The historical LSD corpus combines FAO EMPRES-i CSV events and WAHIS PDF outbreak blocks. The audited raw input comprised 2,407 CSV rows and 682 WAHIS outbreak blocks; a byte-identical duplicate CSV was excluded. The 3,089 retained raw records were normalized and consolidated into 2,587 conservative entries, of which 1,480 were eligible model candidates. The process did not automatically collapse all uncertain duplicate groups. FMD followed a separately specified corpus with 9,526 canonical events, 9,311 meeting confirmed-event inclusion criteria. The disease-specific outputs are not interchangeable because their source and target definitions, available spatial quality and validated artefacts differ.

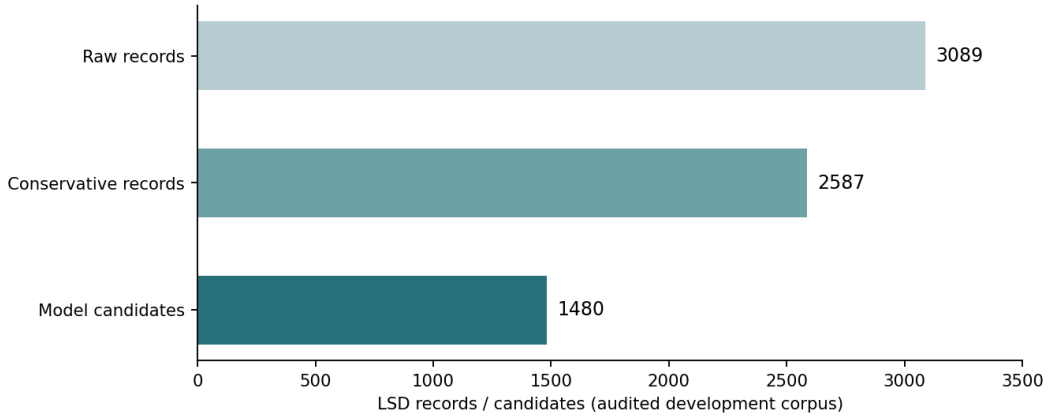


Figure 3.2 LSD corpus processing: raw records, conservative records and model-eligible candidates.

Source: Reconstructed from tracked data-audit counts included in the code-evidence report.

3.2.2 Record structure and quality semantics

Key fields include a source-system identifier, original disease label, normalized disease name, species, event date candidates, report dates, latitude, longitude, approximate-location quality and duplicate-group identity. For research use, the pipeline distinguishes biological event time from operational availability and from storage time. International historical records do not supply exact operational availability for use in a production real-time simulation; the retrospective mode uses a declared event-date proxy. This distinction is not cosmetic: if a report was published after an outbreak, assuming the event date was the information-availability

date could make historical forecasts appear more informed than they would have been at the time.

3.2.3 Spatial quality and geographical scope

Input geographical coordinates are interpreted in WGS84, with GeoJSON transported as longitude followed by latitude. Geodesic computations avoid treating angular degrees as kilometres, while local UTM projected domains support regular grid construction. Records with missing coordinates cannot act as spatial scoring sources. Unknown GPS precision is retained as a quality limitation rather than replaced by invented locations. In the audited LSD conservative corpus, 1,520 entries were recorded with UNKNOWN GPS precision, which is a material limitation for fine-scale farm advice.

3.2.4 Operational clinical and farm context

A second, logically separate data domain is the host ADRS clinical and farm environment. The geospatial component reads vetted diagnostic records, associated farm or herd records and the veterinarian's authorized district through host data adapters. A live confirmed-case marker must not be presented as a model-generated future event. Operational data is subject to authentication and role checks, and the component accesses relevant collections without making itself the source of truth for diagnosis. The research evaluation corpus and operational mapping context are therefore deliberately not merged into a single unqualified training dataset.

3.3 Experimental or study procedure

3.3.1 Reproducible normalization and deduplication

The processing procedure begins by hashing raw input files to detect byte-identical duplicates, parsing source-specific formats and creating normalized records without discarding the original values. Candidate duplicate detection uses hard constraints on country, date and other identity information, followed by conservative locality, spatial and species evidence. Review-level ambiguous groups are retained for inspection rather than automatically merged. Model-candidate construction admits higher-confidence records and documented singletons. The resulting chain preserves source provenance from raw file through deduplicated record, making it possible to trace a displayed historical source to its origin.

3.3.2 Historical forecast origins and leakage controls

A forecast origin is formed at a historical time t0 for a country and disease. Only eligible source records from the accepted preceding window enter its context, and follow-on target events are sought after that origin. The audited LSD protocol uses a fourteen-day historical source window and defines a D1–D7 pooled target-evaluation horizon. Records are assigned to development, held-out or Sri Lanka transfer-study roles. Sri Lanka is excluded from model development; development fitting uses calendar-based folds with a seven-day temporal purge. The protocol thereby separates historical sources from subsequent outcomes and mitigates overlap between training and evaluation windows.

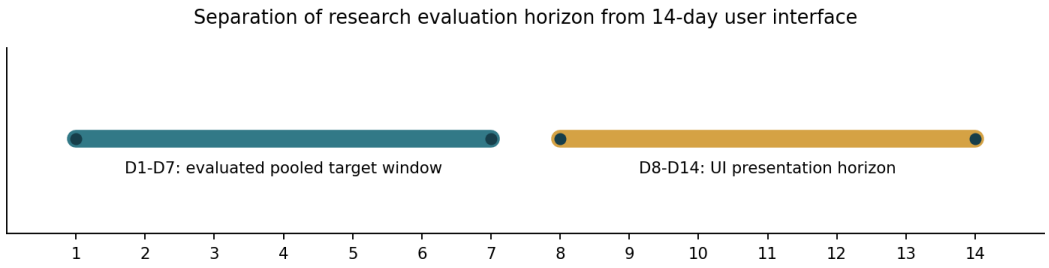


Figure 3.3 Research D1–D7 evaluation horizon versus the fourteen-day user-interface presentation horizon.

Source: Author's interpretation of the audited scientific protocol and recorded demo.

3.3.3 Contextual spatio-temporal clustering

An ST-DBSCAN neighborhood contains an event pair only when their geodesic separation and absolute temporal gap both satisfy the selected bounds. A core event requires sufficient distinct supporting identities, reducing the chance that multiple rows sharing an approximate coordinate artificially form a cluster. Connected core points establish groups, border records are associated through deterministic rules, and isolated events remain explicitly designated as noise. The implementation supplies clustering as descriptive local context; the frozen spatial-ranking model and front-end forecast presentation are not described here as ST-DBSCAN outputs. Formal sensitivity and frozen parameter choices would be necessary to treat one cluster configuration as an evaluated epidemiological conclusion [5].

$$N\varepsilon(i) = \{ j : dgeo(i,j) \leq \varepsilon space \text{ and } |ti - tj| \leq \varepsilon time \} \qquad (3.1)$$

3.3.4 Scientific domain and kernel scoring

For each eligible origin, sources are used to form geodesic spatial components and a bounded grid of approximately 5 km cells. The selected C0 rule is an isotropic exponential kernel with frozen scale 25 km. At cell x, each eligible source s contributes exp(−dgeo(x,s)/25 km); contributions are summed to obtain a relative spatial score. The model has no trained regression weights and does not ingest wind, rainfall or host-density factors in its frozen scoring path. The score is used to rank cells within the constructed domain and is not normalized into an epidemiological incidence probability.

$$C0(x \mid t0) = \Sigma s \in S(t0)\ \exp[-dgeo(x,s) / 25\ km] \qquad (3.2)$$

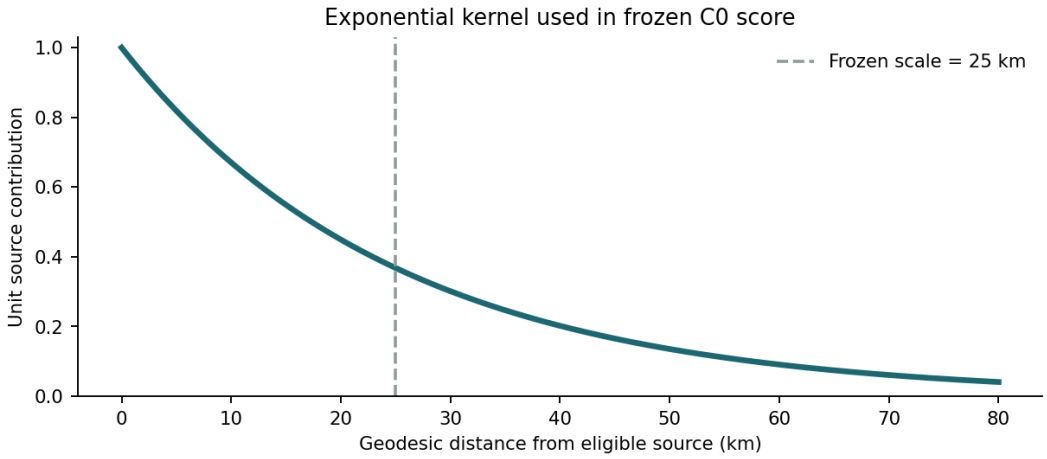


Figure 3.4 Unit-source exponential kernel used by the frozen LSD spatial ranking model.

Source: Analytical illustration of Equation (3.2).

The candidate-development process compared distance-only and host-weighted baselines with exponential and Gaussian kernels over several scales. The frozen reference was the B0 exponential kernel at 25 km. Candidate host-density formulations could not be considered equivalent evaluated competitors because their geospatial coverage was inadequate for eligible comparisons. Similarly, wind-anisotropic candidates had unmet weather-coverage requirements. Scientific selection in this circumstance is based on reproducible eligibility and evidence availability, not on proof that environment-aware physics performs worse.

3.3.5 Geometric direction field

To provide direction context consistent with C0, a cell-local vector is constructed from weighted source-to-cell bearings in an east/north local frame. Its mathematical interpretation is a scaled negative gradient of the C0 spatial surface. The vector is descriptive of the geometry of the source influence field: where influences are balanced, the resultant can be weak; where the contributing directions align, it can be stronger. A displayed angle and directional-clarity descriptor must not be described as a validated pathogen travel bearing or predictive confidence. Its construction was tested against finite-difference identities and audited for mathematical invariants.

$$V(x) = -25\ \text{km} \times \nabla C0(x) \qquad (3.3)$$

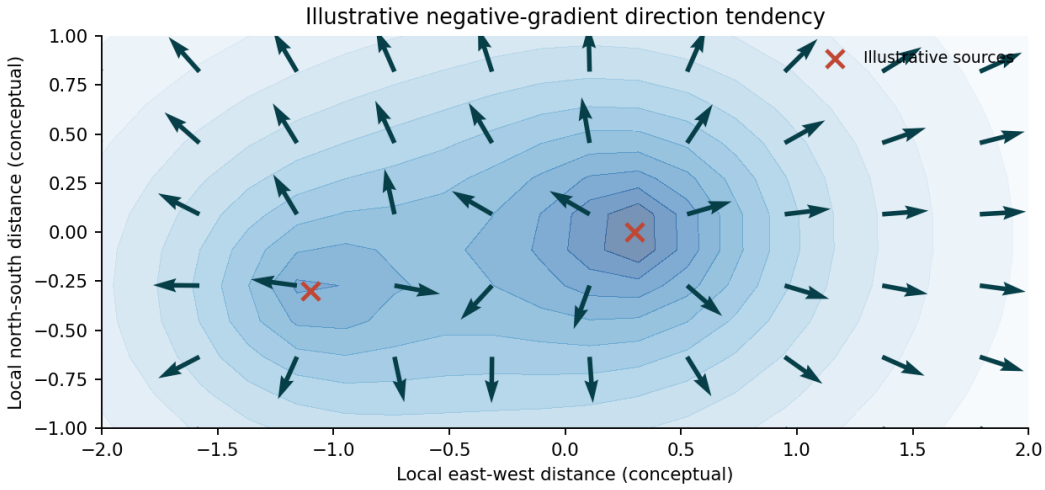


Figure 3.5 Conceptual cell-local negative-gradient arrows derived from a source-contribution surface.

Source: Original explanatory diagram; illustrative source positions, not observed outbreak coordinates.

3.3.6 Historical apparent spread-front and nominal reach

A separate estimator summarizes apparent historical movement relative to eligible source locations. For each source-target relation, geodesic separation is divided by observed lead time; target-event aggregation limits row-level pseudo-replication. The frozen S0 summary is the target-event median, accompanied by a bootstrap interval. Multiplying this estimate by a selected day generates a nominal-reach descriptor. The input sample is limited by a 25 km evaluation envelope, so this value is explicitly scope-conditioned and cannot be interpreted as the speed at which FMD or LSD must spread in any given location. A nominal ring is a contextual graphic, not a quarantine radius.

$$vobs = dmin(target, eligible\ sources) / lead_days \quad (3.4)$$

$$Rnominal(d) = S0 \times d,\ d = 1, \ldots, 7 \quad (3.5)$$

3.3.7 Environmental feature investigation

Meteorological and land-surface modules were investigated as potential extensions: ERA5 temperature, humidity, rainfall and wind; cattle and buffalo density; elevation; land-cover composition; and hydrological proximity. A valid environmental experiment must resolve each factor at or before the origin time, account for geographical coverage and define the transformation used to score each cell. The audited scientific baseline does not use those factors, although data adapters and transformation protocols exist. Treating their mere existence as active forces in the evaluated C0 model would obscure what the reported metrics actually measure.

3.3.8 Interface demonstration protocol

The application demonstration exercises a veterinarian session with national case context, a district-filtered My Area view, timeline play controls, a coloured day-by-day risk-outlook strip, source inspection and an Analysis & Trends screen. A Matara district demonstration illustrates geographic focus and visual changes through fourteen selected dates. For reproducibility, the presentation output and scientific output are documented separately. The fourteen-day ellipse/corridor animation and categorical progression in the audited front-end are generated by deterministic presentation rules rather than the frozen C0 scientific model. The screen recording therefore supports interface and interaction claims only.

3.4 Measures and evaluation metrics

The principal reported LSD scientific outcome is the area-weighted target percentile. It measures where a target cell falls in the ordering of scored grid area, with a midrank rule for ties; a high percentile indicates that the target lies in a relatively high-ranked region of the evaluated area. TOP5 and TOP10 capture consider whether the target falls in the highest 5% or 10% of ranked area, respectively. These are location-ranking metrics; they are not clinical sensitivity, disease-classification accuracy or calibrated probability. Result summaries also report the number of eligible origins, target events and geographic scope coverage.

$$P(target) = 100 \times [\ A(C0 < Ctarget) + 0.5\ A(C0 = Ctarget)\ ] / Atotal \quad (3.6)$$

Complementary engineering measures include consistency of serialized snapshot identity, preservation of longitude/latitude order, response-state correctness, frontend compile/build evidence and tests of authorization boundaries. Direction correctness is evaluated as a structural relation to the C0 gradient, not forecast bearing error. Confidence intervals for the S0 point estimate are generated using resampling; this uncertainty describes the development cohort's apparent rate, not the full uncertainty of a future outbreak trajectory.

3.5 Data analysis

Development results were estimated using expanding calendar-year folds and an origin-balanced perspective to avoid overcounting repeated targets from a single source context. Frozen candidate outputs were then evaluated on held-out historical periods with explicitly recorded coverage restrictions. A separate Sri Lanka case study was used to inspect geographical transfer, but its single evaluable LSD target makes generalization claims inappropriate. Evaluation records were compared against actual source manifests and protocol identities, and aggregate summaries were retained in version-controlled JSON evidence files.

FMD was treated as a different supervised target problem: the occurrence of at least one further eligible event in the D1–D7 window at the origin level. The repository contains a frozen FMD scalar candidate and held-out scoring coverage. Because tracked aggregate artefacts did not supply comparable PR-AUC, AUROC or Brier metrics for final reporting, those values are not inferred here. The narrative concentrates quantitative conclusions on the better evidenced LSD spatial ranking and on validated structural/geometric computations.

3.6 Ethics, privacy, security and responsible AI

The clinical interface restricts district and farm information to an authenticated veterinarian. Historical-source events used in model evaluation are non-personal surveillance records, whereas operational data can identify farms, animal owners and health concerns. The geospatial module accesses host information using read-oriented adapters and enforces authorization on vet-facing context endpoints. Publicly accessible historical scientific routes and development-oriented cross-origin

queries require additional deployment policies when placed in a sensitive production environment.

Responsible display requires explicit separation of observation, model score and simulation. A coloured cell is a relative rank; a geometric arrow is not observed pathogen travel; a nominal ring is not an automatically authorized exclusion zone; and a recommendation card does not substitute for veterinary diagnosis, outbreak confirmation or official animal-movement/vaccination policy. The source code's descriptive and disclaimer labels establish the intended semantic boundary. Any LLM-generated or rule-based intervention suggestion should be presented for professional review and audited before clinical use.

3.7 Reproducibility and research artefacts

The research repository records deterministic parsing and split protocols, mathematical source code, frozen configuration hashes and checkpoint evidence summaries. Raw and processed outbreak files are excluded from version control, so full independent reproduction also requires appropriately licensed input files and the local build manifests. Reproducibility depends on retaining those manifests, exact model candidate identifiers, role assignments and result scripts. The recorded audit used a review branch with a known commit and separately identified uncommitted integration changes; this matters when citing precisely which user-facing build was demonstrated.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Requirements were derived from the approved topic form, the individual proposal and the recorded veterinary walkthrough. They are organized into observed case presentation, historical analysis, veterinarian area operations, explainable visual context and quality controls. The design prioritizes a clinician's sequence of tasks: establish national situation, narrow to the authorized district, inspect the possible spatial implications and evaluate source/farm context before making a professional decision.

ID	Functional requirement	Design response
FR-01	Show the national geographical distribution of relevant confirmed clinical events.	National Map mode with verified marker overlay and selection.
FR-02	Allow the vet to navigate from Sri Lanka to the assigned district.	My District/My Area scope and camera focus.
FR-03	Display an interactive time control and risk-outlook graphics.	Playback, day label, coloured fourteen-step strip and map visual layers.
FR-04	Expose source information underlying a mapped context.	Source and clinical-case popups with traceable metadata.
FR-05	Show local area conditions and relevant farm context.	My Area summary, district indicators and farm-linked operational context.
FR-06	Provide retrospective trends and scientific evidence context.	Analysis & Trends and origin inspection panels.
FR-07	Support authorized response planning.	Explainable recommendations routed for vet confirmation, not automatic enforcement.

Table 4.1 Core practitioner requirements mapped to interface design.

Quality attribute	Requirement interpretation
Accuracy of representation	Never present historical proxies as actual operational timestamps or scores as infection probability.
Security	Vet authentication and district/farm access restrictions are enforced at backend boundaries.
Traceability	Displayed scientific context has origin and snapshot identity; source locations remain auditable.

Quality attribute	Requirement interpretation
Availability	An unavailable scientific result is represented as a controlled state, not fabricated data.
Usability	Consistent map legends, readable timeline states, focused district navigation and accessible popups.
Reproducibility	Frozen research configuration, versioned evidence and separation of presentation from model data.

Table 4.2 Non-functional design requirements.

4.2 Component architecture within the complete solution

The architecture comprises three layers. First, the host ADRS system handles clinical workflows, farm profiles and authenticated veterinarian identity. Second, the individual geospatial backend performs historical research analysis and offers geospatial services, with read-only operational adapters translating host records to map context. Third, the React frontend consumes scientific snapshots and authenticated clinical context to render the National Map, My Area and Analysis & Trends views. Figure 4.1 depicts these layers and the boundary between team data and individual analysis.

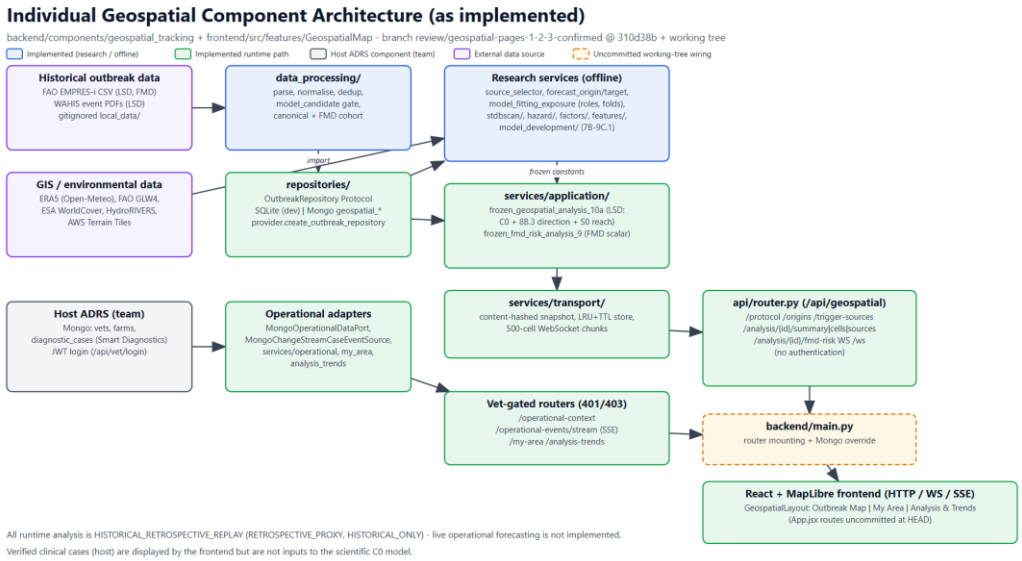


Figure 4.1 Geospatial component architecture and host integration boundaries.

Source: Original diagram based on the geospatial repository structure and host interface contracts.

For a historical origin, the scientific side selects eligible sources, constructs its finite domain and grid, attaches the frozen C0 score, derives geometric context and serializes GeoJSON. For current operations, a vet role-bound endpoint obtains confirmed clinical cases and relevant farms from host collections. Keeping the two pathways independent permits the interface to overlay observed operational events without silently introducing them into a historical model lacking a live availability protocol.

4.3 Design alternatives and decisions

Design decision	Alternative	Selected rationale
Spatial surface	Fully weighted environmental hybrid versus distance-only reference	C0 is reproducible under actual feature coverage; environmental extensions need valid input windows.
Data access	Duplicate host records versus read-only host adapters	Adapters maintain single ownership of clinical and farm data.
Transport	Single large response versus chunked snapshot transport	Content-hashed frames and integrity checks support coherent client rendering.
Display	Only national map versus national map plus district focus	District view supports the veterinarian’s immediate response context.
Future visualisation	Opaque “prediction” layer versus explicitly labelled outlook presentation	Separates interaction prototype from validated numerical forecast.

Table 4.3 Principal alternatives and evidence-led design decisions.

4.4 Detailed design

4.4.1 National outbreak map

After login, the user encounters the Geospatial Intelligence navigation and a national view of Sri Lanka. The interface includes disease and view controls, outbreak markers and selectable map modes. Confirmed operational markers are visually differentiated from historical context. The camera can be adjusted to see the national distribution or focus locally, while source panels expose selected record context. Figure 4.2 is extracted from the application walkthrough and evidences the screen design rather than an epidemiological result.

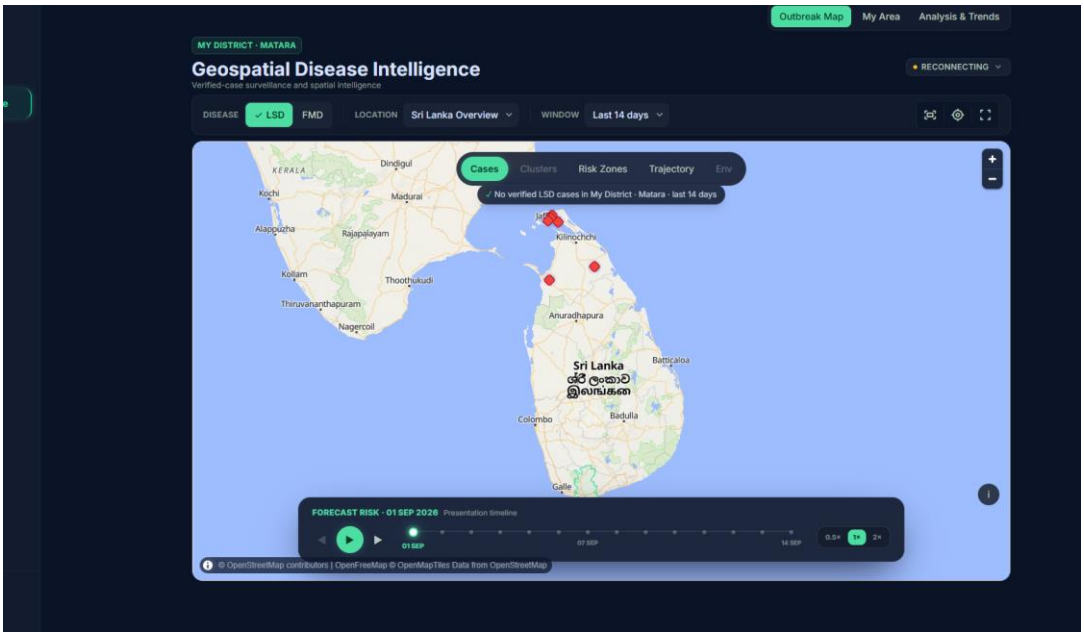


Figure 4.2 National Geospatial Intelligence map showing Sri Lanka overview and observed outbreak markers.

Source: Authentic application capture from Geospatial Individual Thesis Evidence ZIP, 9 October 2026.

4.4.2 Timeline and projected visual layer

The timeline provides play/pause, a current presentation day and a changing map overlay. As the user advances the selected date, source-linked influence corridors and colours change visually to convey a prospective planning horizon. In the Matara demonstration, the sequence spans fourteen dates, with a corresponding qualitative category strip. The layer should be called a projected or illustrative risk-outlook visualization, because the audited animation path and growth parameters are deterministic user-interface rules. Those graphics currently do not establish environmental-force-driven prediction accuracy for fourteen days.

The time control is shared with map and risk-chart presentation states. On each selected day, the displayed source-linked corridor, attention colours and qualitative district risk profile are refreshed together. Verified historical or clinical cases remain marked as observed evidence, whereas projected spatial corridors represent a time-stepped scenario visualization. The frozen C0 research score is a single origin-time surface, so the fourteen frames are not fourteen separately calibrated outbreak-count predictions.

4.4.3 Veterinarian My Area dashboard

My Area is designed around the authenticated veterinarian's registered district rather than a permanently fixed district. The recorded walkthrough uses Matara as its configured demonstration district, showing a locally centered map, a right-hand area intelligence panel, source context and a Future Risk Outlook chart. A district-specific interface is more actionable than requiring a user to repeatedly search the national map. The host's assigned-district data is the appropriate production source for this setting; the demonstrated Matara selection is a known walkthrough configuration and should not be cited as evidence that every veterinarian's district was dynamically verified in production.

The architectural geographic filter is derived from the authenticated veterinarian profile and authorized farm/case relationships. Thus a veterinarian assigned to Galle, Kandy or another district should receive a map camera and context filtered to that district. The October 2026 demonstration applies a Matara-specific view override; this is an implementation configuration used for the recorded walkthrough, not a geographical restriction of the research problem. In the designed general workflow, the signed-in district determines the My Area scope.

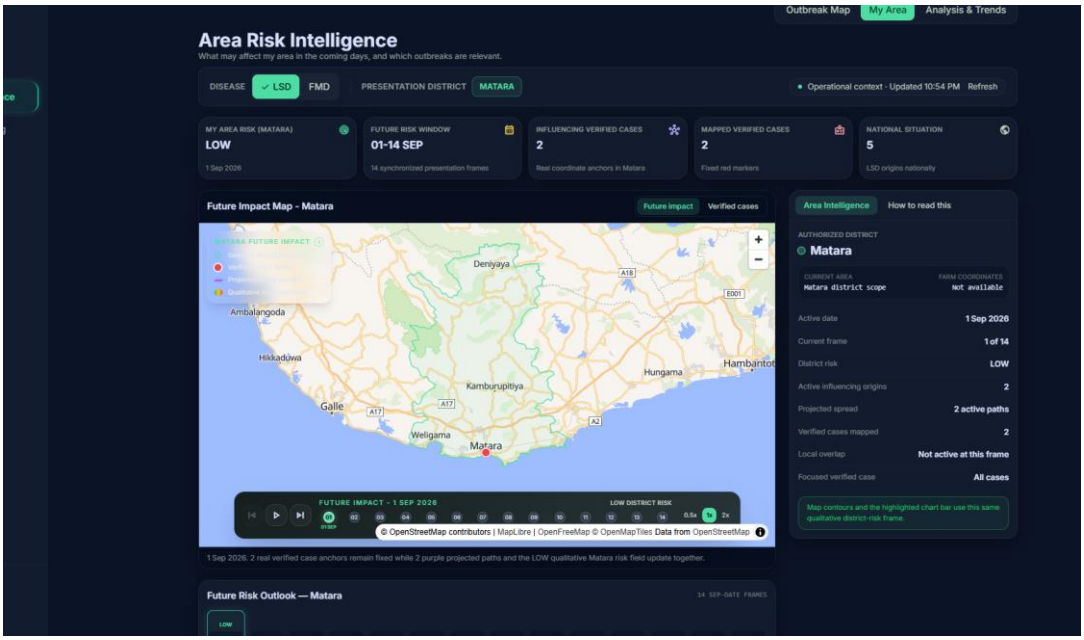


Figure 4.3 Registered-district My Area view illustrated using the Matara veterinary demonstration.

Source: Authentic application capture from Geospatial Individual Thesis Evidence ZIP, 9 October 2026.

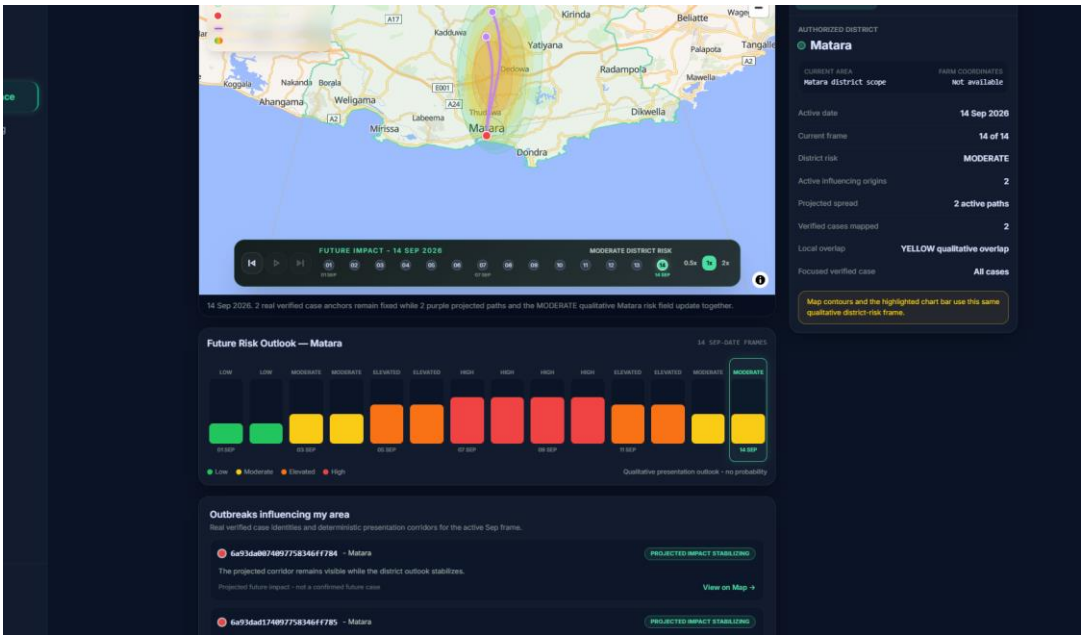


Figure 4.4 My Area qualitative time profile and influencing source list in the Matara demonstration.

Source: Authentic application capture from Geospatial Individual Thesis Evidence ZIP, 9 October 2026.

4.4.4 Source, farm and recommendations logic

Source-detail interaction should distinguish event identity, disease, confirmation status, observed date, geographic coordinate and the relationships used in the analysis. Farm information belongs to an authorized host record, not to a guessed point along an animated line. A local veterinarian may need linked farm identifiers, herd information, observed symptoms, testing status and contact pathways where legitimately available. The recorded current My Area screen shows influencing-outbreak entries and summary context; the wider requested workflow includes richer farm drill-down and AI-assisted recommendations. These are specified here as a reviewed decision-support interface contract and should be accepted only when corresponding audited product evidence is available.

The My Area evidence captures show an influencing-case list with individual source records and a View on Map action that brings the selected event into geographic focus. Farm association is resolved through veterinarian-authorized host records; where a record has no verified farm coordinate the interface cannot legitimately place that farm at an inferred position. The decision-support layer should present contextual actions such as enhanced monitoring, field assessment, movement-control

review or vaccination consultation as veterinary options rather than automatically issued clinical orders.

Recommendations are best modeled as transparent contextual suggestions rather than autonomous commands. For instance, a high-attention region can trigger suggestions to review recent confirmed records, prioritize field investigation, reinforce biosecurity, assess movement controls or consult vaccination guidance. Quarantine and vaccination are disease- and authority-specific decisions and cannot be deduced mechanically from a coloured map. A responsible interface records the evidence source and places the final decision with the veterinarian and applicable animal-health authorities.

4.4.5 Analysis & Trends dashboard

The Analysis & Trends page provides a retrospective complement to the map. It includes disease-context selection, summary indicators and space for historical activity and most-affected-area presentation. Its role is to answer descriptive questions about when records were available and where the observed information concentrates, rather than to generate another live daily forecast. Scientific evidence availability must be accurately communicated if a selected origin has too little data or if an evaluation result cannot be resolved.

The captured Analysis & Trends screen displays current surveillance summary indicators, the disease selector, a recent-activity chart and most-affected areas. Historical model-evaluation and selected-origin analytical services are maintained separately from these operational surveillance counters. This separation prevents the system from implicitly treating a current confirmed-case count as a spatial model forecast.

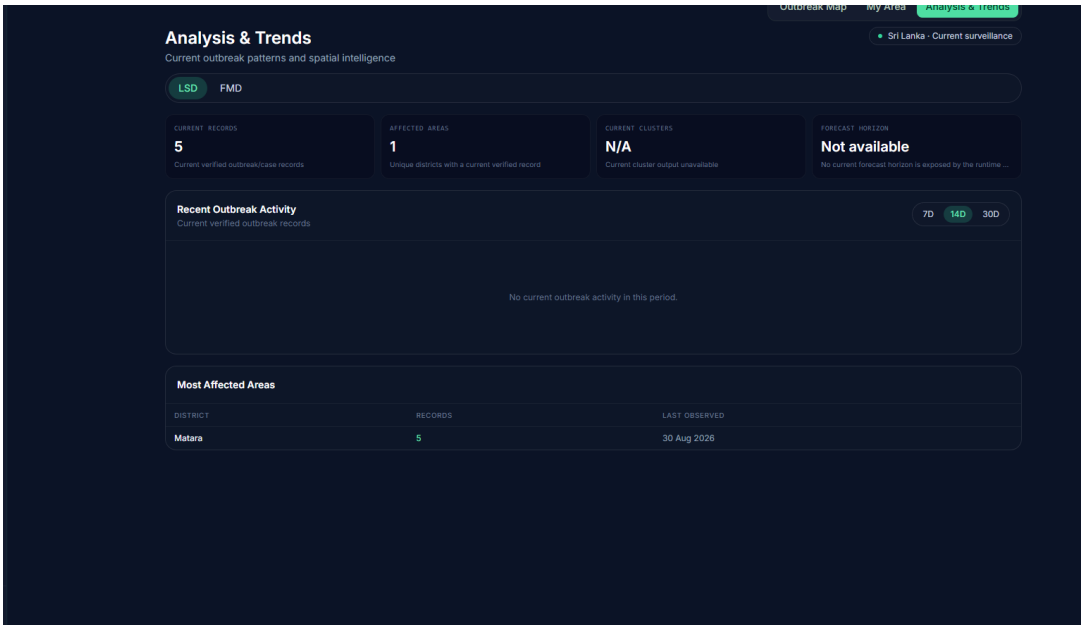


Figure 4.5 Analysis & Trends current-surveillance dashboard and recent outbreak context.
Source: Authentic application capture from Geospatial Individual Thesis Evidence ZIP, 9 October 2026.

4.4.6 Snapshot, WebSocket and state integrity

The scientific services expose HTTP resources for origin listing, source geometry, per-cell data and analysis summaries, in addition to a chunked WebSocket snapshot path. Snapshot content is hashed so clients can reject mismatched or stale frames. The frontend assembles a replacement dataset only after the stream reaches a complete and consistent state. This avoids a frequent spatial visualization error: rendering a new set of cells with old source or rate context and thereby implying a coherent model result that never existed. Figure 4.6 abstracts the resulting request/transport flow.

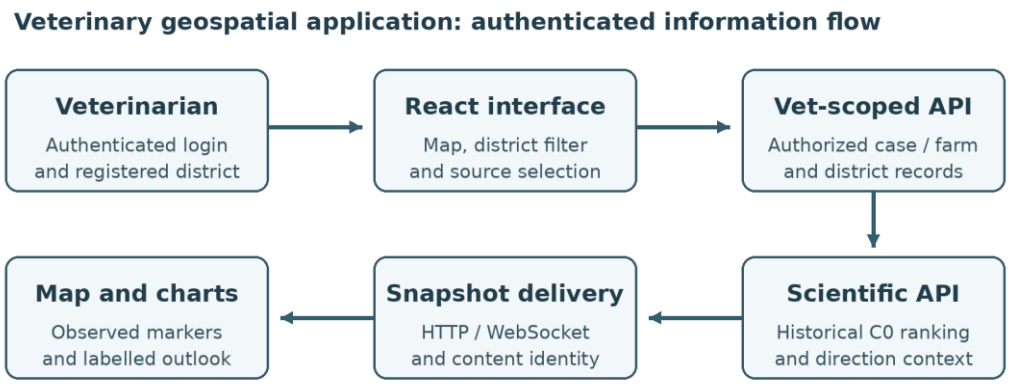


Figure 4.6 Veterinary authentication, operational case context and scientific snapshot delivery.

Source: Author’s reconstruction of authenticated host and historical research API boundaries from source code.

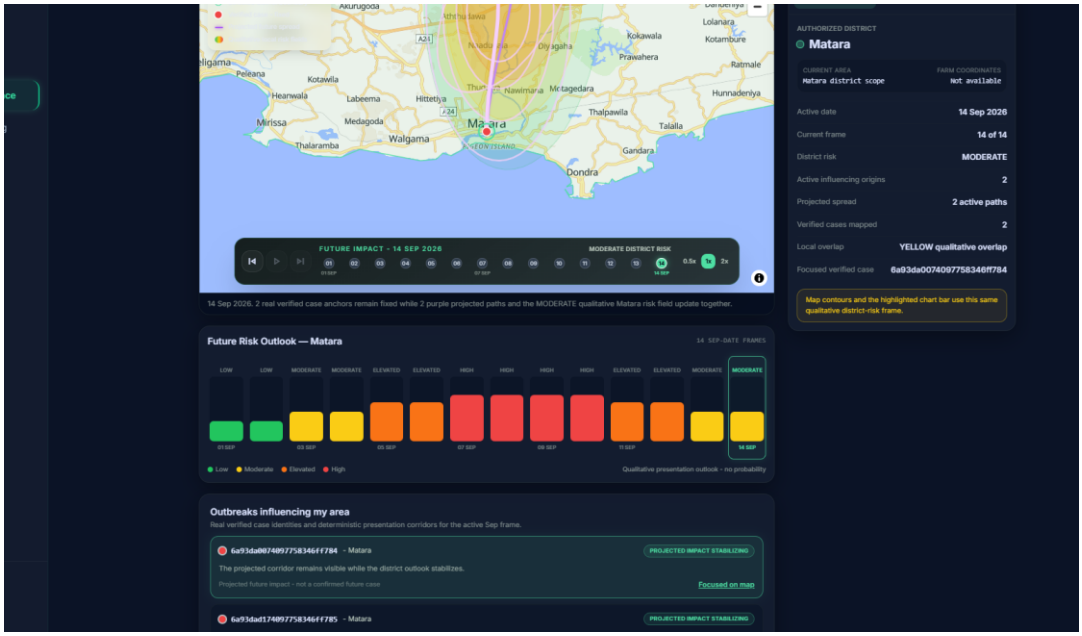


Figure 4.7 Selected verified-case source and the map-focus interaction in My Area.
Source: Actual interface screenshot from the geospatial evidence archive (9 October 2026).

The source inspection controls provide a traceable link between a verified case identifier and its visible map position.

4.5 Standards and professional practice

Geospatial responses follow EPSG:4326 for interchange, with GeoJSON coordinate order [longitude, latitude], while scientific grids use localized projected coordinates for distance-based geometry. API response validation, authorization, source identity and immutable snapshot hashes support stable communication. Tables and charts include labels, numeric units and source descriptions. The interface should use clear contrast for colour bands and retain textual risk-level equivalents, because red/orange/yellow/green alone can be misread or inaccessible. Professional practice further requires differentiating computational evidence from clinical judgment.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development environment and technology stack

The component is implemented with a Python geospatial backend and a JavaScript/React frontend. FastAPI provides typed HTTP and WebSocket interfaces. Scientific computation uses geodesic and raster/vector libraries to build domains and scores. MongoDB integration is available through host-connected adapters, while SQLite provides a reproducible development repository. React, Vite and MapLibre GL JS support the map and page-level interaction state. These technologies were identified from the code-audit evidence; the audited environment version is not presented as a formal minimum supported deployment specification.

Layer	Technology / responsibility
Backend	Python; FastAPI; Pydantic schemas; scientific and operational routers
Spatial and feature processing	pyproj geodesics; Shapely geometry; Rasterio for reference rasters
Machine learning experiments	scikit-learn for FMD experimental candidate models
Storage	MongoDB host collections and geospatial read repositories; SQLite local research fallback
Frontend	React, Vite, MapLibre GL JS, client state / adapters
Communication	GeoJSON over HTTP; content-hashed chunked WebSocket snapshots; SSE event updates
Verification	Pytest contract/invariant tests and Vitest React/adapter tests

Table 5.1 Implementation technology stack.

5.2 Component implementation

5.2.1 Dataset and provenance modules

Parsing modules accept source-specific CSV and WAHIS structures. Normalization generates stable record identities and isolates epidemiologically meaningful dates from file-import times. Deduplication logic uses deterministic high-confidence matching and explicit review groups. Canonical builders then emit analysis-ready candidate datasets and source manifests. The separation of each transformation into an independently testable module is important because wrong deduplication can

contaminate clustering, distance fields and evaluation metrics before any map is rendered.

5.2.2 Scientific services

The scientific service layer implements source eligibility, historical forecast origins, target-event grouping, spatial domain construction, rank scoring and frozen candidate registries. Hazard-kernel primitives and environment adapters are retained as extensibility work, but the evaluated C0 runtime path is specifically an exponential distance-score calculation. A direction service converts cell-local vectors into bearings, and nominal reach is served alongside explanatory status labels. This modular design allows interface functions to remain stable while a later scientifically validated model is substituted behind an explicit versioned protocol.

5.2.3 API resources

Endpoint family	Illustrative purpose	Authorization / semantics
GET /api/geospatial/origins	List historical forecast origins by disease/country	Historical research resource; data is retrospective
GET /api/geospatial/analysis/{id}/summary	C0 summary, rate context, nominal reach	Historical score; not disease probability
GET /api/geospatial/analysis/{id}/cells	GeoJSON score cells and bearings	LSD spatial model path
GET /api/geospatial/analysis/{id}/sources	Eligible historical source records	Provenance linked to selected origin
GET /api/geospatial/analysis/{id}/fmd-risk	Separate FMD origin-level score	Not equivalent to an LSD trajectory
WS /api/geospatial/ws	Chunked scientific snapshot delivery	Integrity and snapshot identity controls
GET /api/geospatial/operational-context	Relevant verified cases and farms	Vet authentication and host context
GET /api/geospatial/my-area	District-specific operational/research context	Vet role and district scope

Endpoint family	Illustrative purpose	Authorization / semantics
GET /api/geospatial/analysis-trends	Historical trends and origin analytics	Vet role; research/descriptive
GET /api/geospatial/operational-events/stream	Verified-case update stream	Vet-authenticated SSE

Table 5.2 Selected geospatial service interfaces and their interpretation.

5.2.4 React interface and map-state separation

The front-end component tree separates page orchestration from data adapters, map geometry and presentation functions. The National Map can combine outbreak markers, selectable origins and map modes. My Area binds district context, local map view, a forecast-outlook strip and related-source cards. Analysis & Trends provides historical summary panels. Specialized adapters handle source geometry, disease capabilities, date formatting and current-context mapping, while snapshot-state logic prevents stale chunks and cross-request contamination. Consistent scientific labels are as important here as the rendering library, because a screenshot can be persuasive even when the underlying layer is illustrative.

5.2.5 Presentation projection versus numerical model

The development front end implements a visually continuous fourteen-day animation by deterministic propagation and categorical progression rules. In the recorded example, coloured corridor geometry progresses across the Matara district with the playback date. However, these paths are not derived from a fitted environmental vector-field forecast. Conversely, the real C0 scientific outputs are a static origin-time grid score and derived geometric direction field, with the S0 nominal reach context available for seven days. Software design therefore uses separate objects and semantic labels for observed source, evaluated cell, historical reach and illustrative future overlay.

5.3 Integration with the other components

Integration begins at the vet portal. The shared host controls login, role and district identity, farm context and diagnostic confirmation. Once a clinical case is marked verified, the geospatial adapter can surface it as an observed operational marker.

Mongo change-stream / SSE mechanisms support event-based view updates and ordinary HTTP reconciliation supports recovery when a stream is interrupted. The geospatial component reads those records rather than editing the underlying diagnosis. Thus the relationship is data consumption and visualization, not scientific certification of the diagnosis algorithm.

The reviewed working branch contained integrated backend and frontend host wiring alongside uncommitted modifications, while code-evidence summaries include earlier scientific checkpoints from a component branch. The thesis treats the demonstration as an observed interface artefact and does not assert that all integration code is part of the repository's default branch or a production deployment. Recording the branch and commit of a final submission build is essential for reproducibility. The inspected review commit was 310d38be15c8cf307c454cde69a21bae20d87e72; changes not committed at the time of audit cannot be assumed available in that commit alone.

5.4 Verification and software testing

The verification strategy is intentionally layered. Data tests protect identity, duplicate rules and event chronology. Geometry tests cover projections, coordinate order, geodesic distances and gradient identities. API tests examine response schemas, error-state mapping and veterinarian role guards. Transport tests check frame order, incomplete ranges, duplicate chunks, request identities and equivalence of HTTP and WebSocket snapshots. Frontend tests cover page wiring, selectors, status labels and visual layer adapters. Functional page demonstrations provide additional interaction evidence but do not replace scientific validation.

Evidence group	Recorded audit evidence	Interpretation
Backend inventory	2,751 pytest cases collected; collection only	Coverage inventory, not a current complete pass
Historical backend checkpoint	1,636 passed, 0 failed in recorded 11A checkpoint	Historical checkpoint test run
Frontend checkpoint	31 passed at 11A; 118 passed at later 11B focused run	Recorded subsets, not current complete run
Direction structural audit	560,853 scored cells; zero invariant failures	Mathematical/structural consistency

Evidence group	Recorded audit evidence	Interpretation
API smoke test	One origin, 88 cells, 0.18 s	Single-origin structural test, not load benchmark
Veterinarian authorization	401/403 response tests in route factories	Contract-level access verification
Live outcome validation	No independent field-outcome study supplied	No clinical forecast accuracy claim

Table 5.3 Testing evidence and what it establishes.

5.5 Deployment and operation

The application is demonstrated through a local development environment using a browser-addressable veterinarian route. Scientific snapshot endpoints can be served from a FastAPI process, while the React client runs with application proxy or configured API base path. The map may use an external base-map style or a token-free fallback. Effective operation depends on correct host database connectivity, JWT configuration, authorised district assignments, available reference data and consistent model artefacts. The evidence does not establish a production roll-out, field uptime, load capacity or automatic outbreak response, and these are not represented as completed deployment results.

Operational readiness also requires a defined privacy policy for farm coordinates, audit logging, rate-limits, secure transport, dependency patching, observability and routine science-version checks. A district-level cache may improve performance, but it must not be used to bypass permissions or expose another veterinarian’s information. Scientific model versions and user-interface projection versions should be displayed or logged independently for future evaluation and incident review.

6 RESULTS AND VALIDATION

6.1 Research results

The principal quantitative result is the retrospective LSD relative spatial-ranking evaluation of the selected C0 distance-kernel model. On development origins contributing evaluable targets, the mean area-weighted percentile was 62.03; the held-out result was 73.85 with a recorded origin-level 95% interval of 70.51–77.11. The held-out coverage was conditional on a 25 km spatial envelope: 323 of 588 target events fell inside scope. These results indicate where target events ranked within the evaluated area, not the probability that disease will arise at those points. Part of the held-out cohort was exposed before final selection and that limitation must remain associated with the result.

Evaluation	Origins / targets	Metric	Recorded result
Development (7B)	277 contributing origins; 1,302 targets	Mean area-weighted target percentile	62.03; interval 59.77–64.39
Held-out (7D)	126 contributing origins; 323 in-scope targets	Mean area-weighted target percentile	73.85; origin interval 70.51–77.11
Held-out (7D)	Same evaluation scope	TOP5 / TOP10 capture	0.174 / 0.292
Sri Lanka (7E)	5 origins; 1 evaluable target	Target percentile	61.10; descriptive single target
Direction (8B)	560,853 cells audited	Structural invariant failures	0
Rate estimation (9B)	371 target events	S0 median, bootstrap 95% interval	3.946 km/day [3.549, 4.343]

Table 6.1 Primary research evidence from tracked checkpoint summaries.

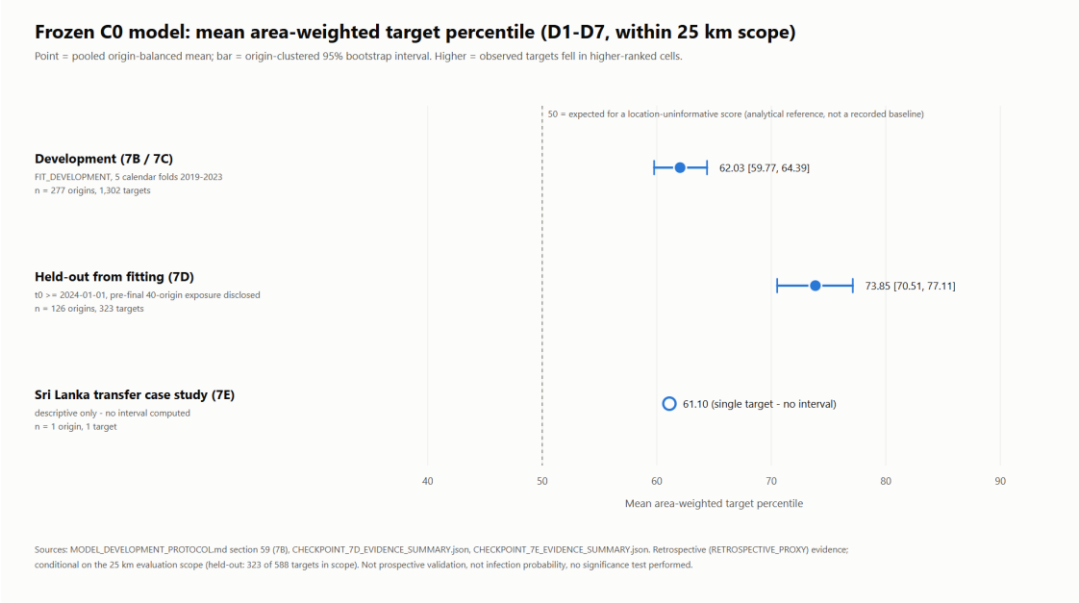


Figure 6.1 Development, held-out and Sri Lanka LSD area-weighted ranking summaries.

Source: Chart reconstructed from tracked checkpoint-level aggregate values; cohorts have different evaluable scopes.

The held-out ranking should not be read as externally verified prediction accuracy. About 55% of held-out targets were within the 25 km domain, Sri Lanka supplied only one evaluable LSD target, and the result reports geographic ordering rather than binary case-classification outcomes. The selected model is therefore most appropriately described as an interpretable spatial attention-ranking baseline. Stronger local conclusions would require more confirmed, geographically precise Sri Lankan targets and a prospectively defined evaluation scheme.

6.2 Product and technical results

6.2.1 Direction and rate outputs

The recorded direction audit found zero structural invariant failures over 560,853 scored cells, supporting geometric consistency between cell vectors and the underlying C0 surface. It does not measure the bearing of real future outbreaks. The historical rate estimator S0 was 3.946 km/day, with a bootstrap interval from 3.549 to 4.343 km/day. The nominal radius obtained by multiplying S0 by days one through seven ranges from approximately 3.95 km to 27.62 km. Day seven exceeds the 25 km selection envelope, another reason not to treat these circles as reliable containment boundaries.

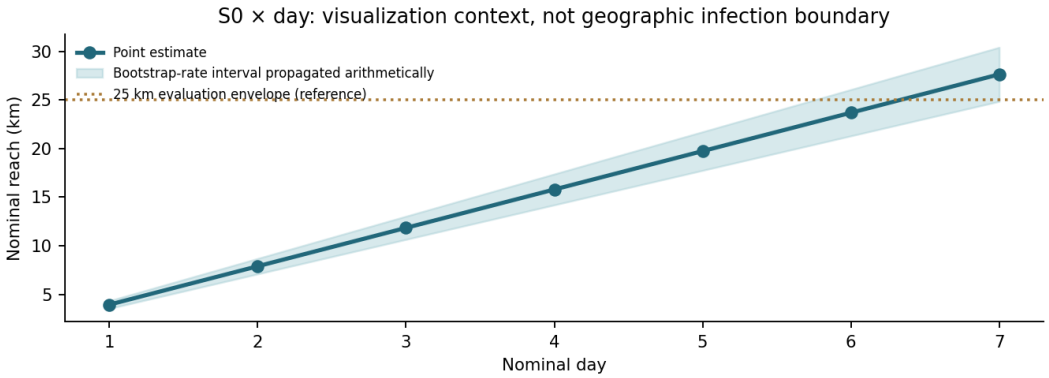


Figure 6.2 Seven-day nominal reach obtained from the historical S0 estimate.

Source: Calculated from recorded S0 estimate and bootstrap interval; not a calibrated infection frontier.

6.2.2 National map and confirmed-case visualization

The recording demonstrates a working navigation from the veterinary portal into Geospatial Intelligence with a Sri Lanka map, markers and map-mode controls. The map interaction establishes that observed/outbreak symbols and geographical focus can be presented in the product shell. It does not by itself verify the provenance of every marker displayed in the video, so the architecture remains anchored to the independently inspected verified-case read interface. The technical contribution is a common geographic surface for source inspection, district focus and visual context.

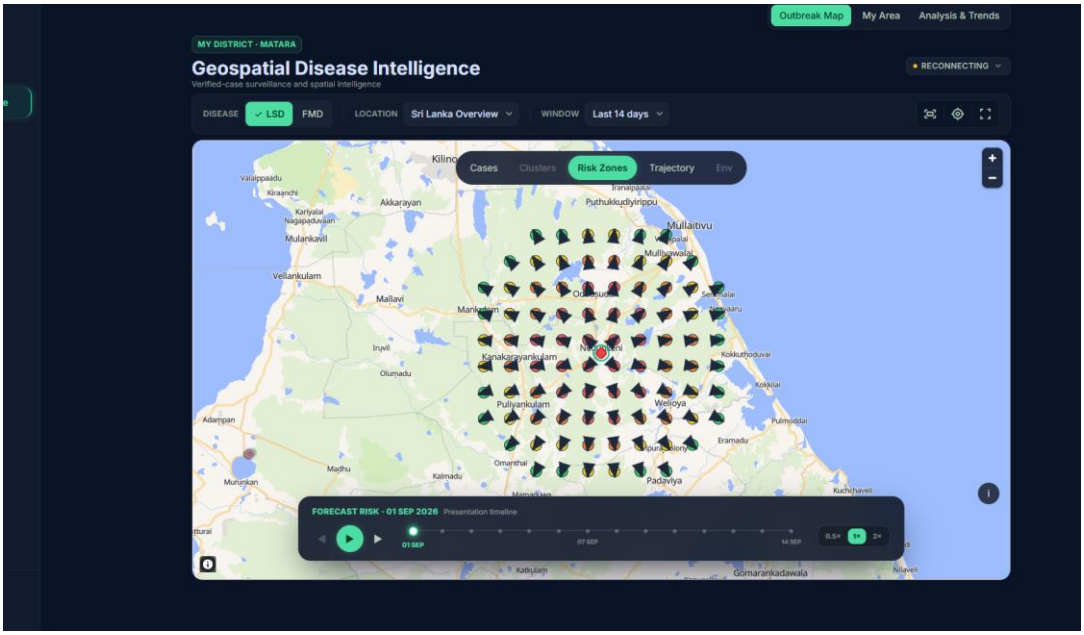


Figure 6.3 National map risk-mode view displaying historical C0 relative spatial score cells.

Source: Authentic application capture from Geospatial Individual Thesis Evidence ZIP, 9 October 2026.

6.2.3 Registered-district My Area and fourteen-day outlook

In the recorded My Area sequence the map camera focuses on Matara, which is the demonstration account's registered district; a right panel identifies the authorized district context and a sequence of fourteen qualitative day states is visualized. The bar colours progress from lower to elevated contextual levels before returning toward moderate levels. An outbreak-influence list includes source identifiers and a View on Map action. This constitutes concrete interface evidence for the requested workflow of examining a district, seeing projected spatial influence and navigating back to source context. The fourteen colours and corridors remain presentation values rather than validated daily incident forecasts.

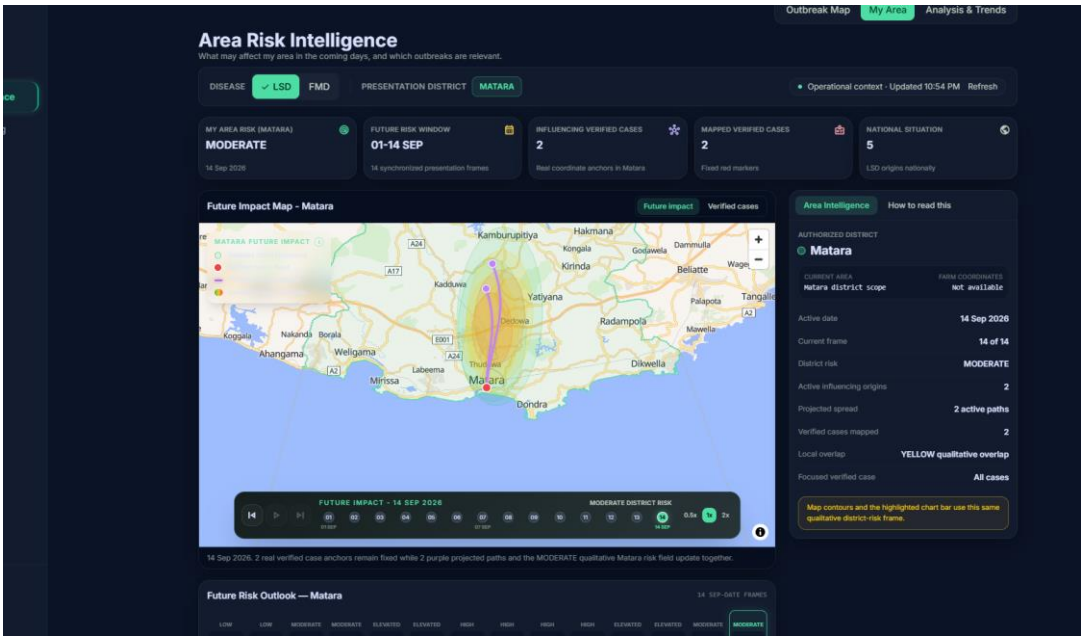


Figure 6.4 District-oriented My Area interface with Day 14 outlook visualization.

Source: Authentic application capture from Geospatial Individual Thesis Evidence ZIP, 9 October 2026.

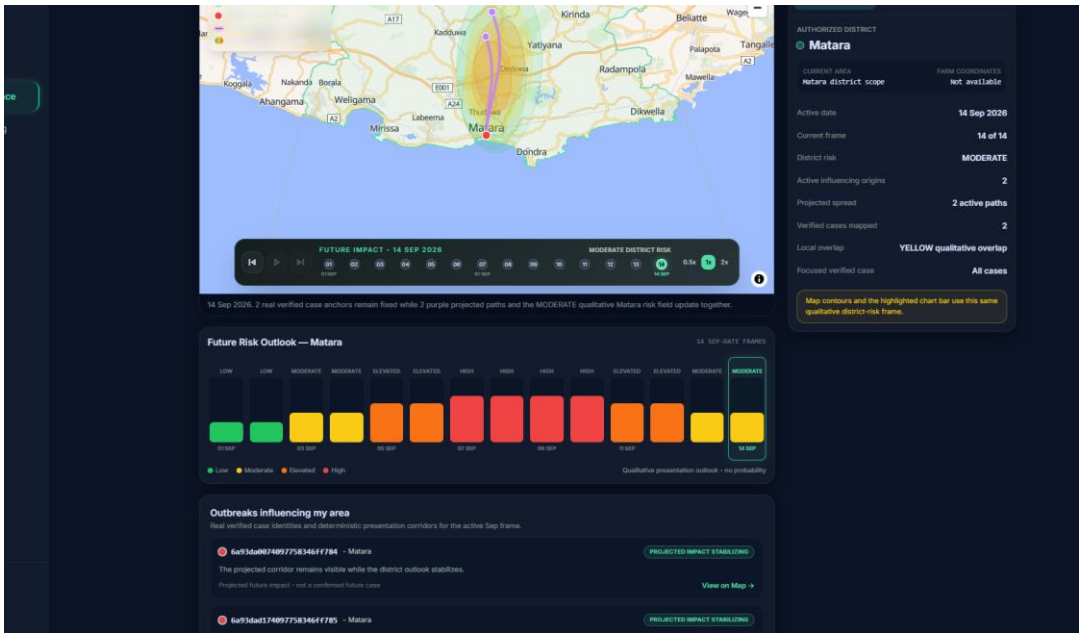


Figure 6.5 Fourteen-day Future Risk Outlook qualitative profile and influencing-case panel.

Source: Actual interface screenshot from the geospatial evidence archive (9 October 2026).

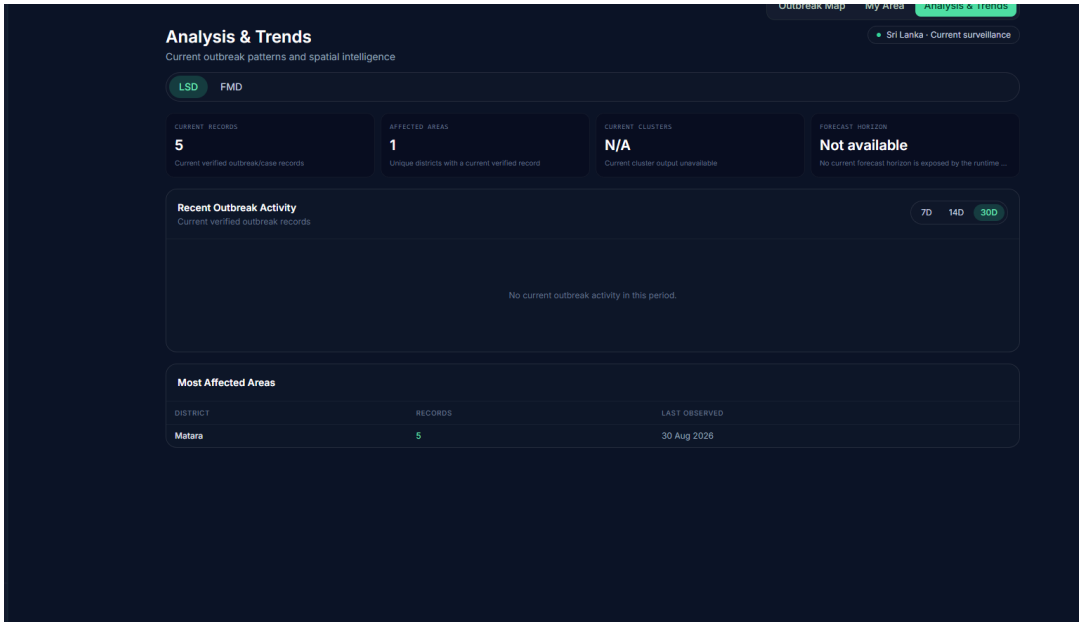


Figure 6.6 Historical disease-surveillance activity presented in Analysis & Trends.

Source: Actual interface screenshot from the geospatial evidence archive (9 October 2026).

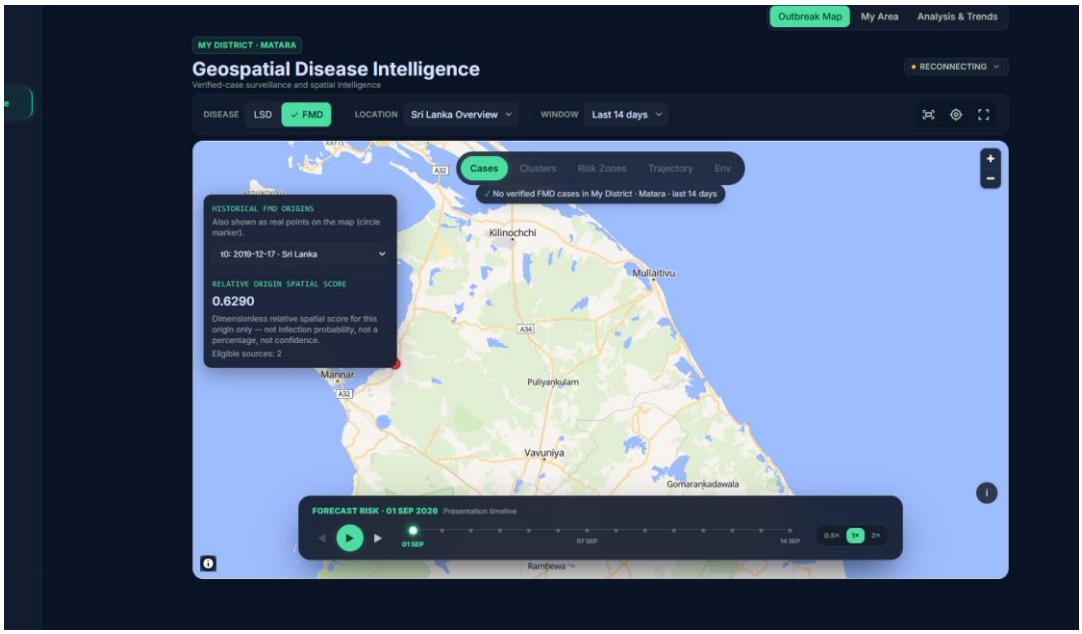


Figure 6.7 National disease-context switch and FMD-oriented surveillance view.

Source: Actual interface screenshot from the geospatial evidence archive (9 October 2026).

The LSD grid-based spatial score and the FMD scalar-origin score have different statistical meanings and are presented with disease-specific labels.

The Analysis & Trends view supports historical and current surveillance interpretation through visible counts and activity windows.

The bars are categorical time-stepped presentation values; the underlying selected source list is independently grounded in verified cases.

6.2.4 Historical trends and inspection

The Analysis & Trends page in the recorded application provides a distinct destination for descriptive metrics rather than overload the operational map. The screen shows top-level indicator panels and an activity/affected-area structure. Its usefulness depends on correct binding to historical versus clinical context and on avoiding default-zero values that imply a genuine absence of cases. The research code includes a historical analysis-trends service with origin-scoped data and explicit unavailable statuses; the integrated page demonstrated in the audit also incorporates verified clinical summaries. These sources should be attributed when final screenshots are captured.

6.3 User or stakeholder validation

The current evidence supports application walkthrough observations, not a structured field usability trial with veterinarians. No signed feedback study, questionnaire sample, task-completion analysis or controlled time-to-decision comparison was supplied. Therefore perceived usability improvements, reduced outbreak response time and clinical effectiveness are stated as design motivations rather than measured outcomes. A field trial can later ask veterinarians to locate confirmed cases, explain the difference between observed and projected layers, inspect influencing sources, and decide whether a professional response is justified.

6.4 Integration results

The existing service contracts support a division among historical source lists, static C0 grid snapshots, selected-origin direction/reach context, authenticated operational case data, district context and event-stream transport. Structural smoke records document a scientific snapshot of one origin with 88 cells and one source; separate transport tests check identity and reconstruction. Review of the working tree additionally shows host wiring for the three-page interface. These observations support feasibility of component assembly under the demonstrated environment, but they are not a measured deployment availability or large-scale capacity result.

6.5 Summary of findings

The most defensible research finding is that a provenance-aware, leakage-controlled historical pipeline can generate interpretable spatial rankings with recorded retrospective target-percentile results. A second result is that direction fields and apparent-rate summaries can be exposed with mathematical transparency and bounded interpretation. The engineering result is a veterinarian-oriented web workflow joining national and local clinical context to scientific inspection and visual planning. The results do not establish environmental-force prediction of the next fourteen days, automatic clinical diagnosis or effectiveness of quarantine/vaccination advice; these remain questions for separate scientific and operational evaluation.

7 DISCUSSION

7.1 Interpretation of findings

The frozen distance-only C0 kernel performed meaningfully as an evaluable ranking artefact. It produces a continuous field and a deterministic cell ordering without requiring complete weather, land-use or livestock-density features. This is a practical advantage when surveillance data are sparse. Its apparent held-out percentile exceeded the development summary, but the cohorts, domain coverage and pre-final exposure caution against inferring model improvement or geographic generalization. An increased percentile can reflect differences in target distribution, source density or evaluation eligibility rather than better representation of disease processes.

Direction and nominal-reach displays illustrate a second central interpretation challenge. An arrow that is mathematically consistent with a spatial score field can still be wrong about physical transmission; a ring calculated from an historical median can still extend beyond the conditions under which the median was measured. The value of the component is increased by exposing the construction and boundaries of these descriptors, enabling a veterinarian to treat them as investigative context rather than laboratory-confirmed outbreak forecasts.

7.2 Comparison with literature, baselines and existing solutions

Compared with global WAHIS and EMPRES-i data portals [3], [4], the individual product places an emphasis on local practitioner interaction and the integration of clinical markers with spatial analysis. Compared with the core ST-DBSCAN literature [5], its clustering implementation provides a configurable context layer rather than treating every cluster as an observed transmission chain. Compared with kernel-estimation theory [6], the frozen C0 surface is a discrete spatial rank score built from an exponential distance kernel; it is not a statistically calibrated local incidence density estimate. The distinction is important when users interpret coloured map overlays.

Candidate studies in the repository also clarify why a simpler baseline was selected. Host-weighted variants were ineligible where the required livestock-density reference coverage was missing, and wind anisotropy could not be compared

consistently across needed historical folds. The evidence does not support a conclusion that those concepts are biologically unimportant. A scientifically justified environmental-vector model should revisit them after local coverage and pre-origin timestamp availability have been independently audited.

7.3 Research contribution

Research contributions comprise four connected results. The first is a source-provenance and deduplication pathway that preserves uncertainty instead of silently forcing ambiguous source identities. The second is outbreak-triggered retrospective replay with leakage firewalls and an origin-balanced geographic rank evaluation. The third is a transparent spatial-scoring baseline with an analytic geometric vector derivation. The fourth is a scope-conditioned historical apparent-rate estimate that explicitly separates nominal visualization reach from actual disease front speed. The combination provides a reproducible model interpretation pathway suitable for cautious veterinary spatial intelligence.

7.4 Product contribution

The product contribution is a coherent three-page workflow. The National Map establishes situational overview and confirmed-case inspection. My Area answers district-specific questions through map focus, a fourteen-step outlook visualization, influencing source records and area intelligence. Analysis & Trends provides retrospective context without requiring practitioners to interpret internal data tables. The host integration uses authentication-aware operational adapters while retaining a separate historical scientific layer. Future additions such as richer farm inspection and recommendation engines fit naturally into this navigation but should be assessed for evidence quality before clinical deployment.

7.5 Limitations and threats to validity

Threat	Effect on interpretation	Mitigation / reporting response
Historical reporting date proxies	Available-at-t0 information may not match real-world availability	Explicit retrospective provenance and no prospective accuracy claim
Sparse Sri Lanka LSD records	Transfer case cannot support national effectiveness inference	Report n=1 evaluable target; collect local field data
Domain-constrained percentile	Targets beyond 25 km are excluded from ranking evaluation	Report in-scope denominator and avoid unconditional statements

Threat	Effect on interpretation	Mitigation / reporting response
Unavailable environmental predictors	Frozen model omits potential disease drivers	Document adapters and coverage preconditions; do not attribute results to them
Unfrozen ST-DBSCAN settings	Cluster conclusions depend on parameter choice	Present method as contextual and test sensitivity
Demonstration future animations	Attractive visuals may be mistaken for fitted forecasts	Separate projected interface illustrations from C0 evaluation
Working-tree integration evidence	Final deployed code provenance is not established by screenshots alone	Pin audited build version and retain test logs
No independent end-user study	No quantified usability or decision outcome effect	Plan veterinary task-based field evaluation

Table 7.1 Threats to validity and conservative interpretation.

A particular visual threat is semantic overclaiming. Colours such as red and orange communicate urgency even without numerical calibration. A fourteen-day time slider suggests day-specific precision that the validated D1–D7 pooled target study does not establish. Moreover, the current Matara demonstration is not proof that the assigned-district logic adapts correctly for all users. These distinctions should be presented within the manuscript and, where practical, in the interface legend and exported reports.

7.6 Implications

For veterinary users, this research motivates a map-based review workflow that begins with observed and confirmed information. For researchers, it shows the importance of explicit timestamp semantics and quality-controlled event identifiers in retrospective outbreak modelling. For software teams, it shows why immutable scientific snapshots, role-specific context adapters and independent presentation-state tests are useful in decision-support applications. For policy-makers, the work is an exploratory geospatial intelligence component, not a validated replacement for official surveillance, epidemiological investigation or government-mandated control measures.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The potential users are district veterinarians, government animal-health officers, extension services and approved livestock-health organizations. The value proposition is quicker access to a coherent geographical view of confirmed cases and associated research context, reducing the need to manually reconcile isolated records, farm lists and maps. The ability to focus on an assigned district supports routine surveillance and more targeted professional investigation. The study has not quantified time savings, adoption willingness or clinical outcomes, so this value proposition remains a product rationale rather than a proven economic benefit.

8.2 Market and competitive position

Global animal-health services already offer event dashboards and validated reporting tools [3], [4]. ADRS should not attempt to replace official WOAH reporting. Its proposed differentiation is the local integration between authenticated veterinary workflows, farm-oriented records and contextual spatial analysis. Commercial comparison would require interviews with Sri Lankan veterinary stakeholders, examination of DAPH procurement and technology constraints, and a privacy-compliant assessment of competing software. Without those studies, market-share, revenue or cost-reduction projections would be speculative.

8.3 Business or sustainability model

A public-sector or institutional deployment model is plausible: software can be hosted by an authorised veterinary organization with documented data stewardship, maintenance arrangements and incident-response responsibilities. Sustainability costs would include secure hosting, storage, mapping services where applicable, software upgrades, veterinary user training, local data acquisition and periodic scientific recalibration. An open and modular implementation can reduce dependence on a single map vendor, but maintenance of source mappings, disease-specific rules and information security remains a continuing operational requirement.

Any future AI recommendation capability should be governed through explicit model cards or rule descriptions, audit logs, versioned veterinary policy content and review by domain professionals. Automatic animal-movement restrictions or

vaccination mandates based solely on an uncalibrated geospatial rank would create both scientific and regulatory risk. A safer service approach is to provide triage suggestions and contextual evidence, leaving intervention decisions with responsible licensed personnel.

8.4 Scalability, intellectual property and compliance

Scaling from a local demonstration to a province- or country-wide operational environment requires a secure backend service, authenticated event streams, auditable record permissions, updated reference data and repeatable model deployments. Performance claims should be established using representative burst loads and realistic geographical cell counts. Intellectual-property and licensing review must consider public map tiles, geospatial data, open-source packages, imported surveillance source terms, team code ownership and institutional policy. Farm coordinates and clinical records need purpose-limited access and retention controls.

8.5 Contribution to the Sustainable Development Goals

The proposed veterinary intelligence workflow aligns primarily with SDG 2 (Zero Hunger) by supporting livestock health and food-system resilience, SDG 3 (Good Health and Well-being) through improved animal-health surveillance interfaces, and SDG 8 (Decent Work and Economic Growth) through the aim of protecting rural livestock livelihoods. These are intended areas of contribution, not measured SDG outcomes of the current prototype. Verification would require deployment studies assessing outbreak-report timeliness, veterinary response and tangible animal-health benefits.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This individual research designed and examined a geospatial intelligence component of ADRS that links imperfect historical livestock disease reports with practitioner-facing spatial visualization. The work addresses data provenance, spatio-temporal eligibility, geodesic spatial ranking, geometric direction context, descriptive reach estimation and veterinarian-focused map interaction. The strongest numerical evidence comes from a frozen LSD C0 spatial rank model evaluated on historical development and held-out origins; the strongest engineering evidence comes from source-processing contracts, structural direction checks, service serialization and the recorded three-page interface walkthrough.

The central conclusion is that responsible geospatial decision support requires a clear distinction between observed confirmed cases, retrospective spatial-model context and future-oriented user-interface illustrations. The product can help organize what a veterinarian sees and investigates without asserting that every displayed path is a scientifically calibrated forecast. This evidence-preserving separation is a practical requirement for moving from an academic demonstration toward a trusted surveillance platform.

9.2 Achievement of objectives

Objective	Evidence-based achievement
O1 — Data quality	Historical LSD/FMD canonicalization, source provenance, temporal rules and conservative duplicate handling established.
O2 — Spatio-temporal research	Geodesic source selection, descriptive clustering services and an evaluated C0 relative spatial-ranking method.
O3 — Geometric context	Mathematically checked local direction geometry and a scoped historical apparent-rate / nominal-reach descriptor.
O4 — Veterinary interface	National Sri Lanka map, authenticated district-oriented My Area workflow, qualitative fourteen-day outlook, verified-source inspection and Analysis & Trends.
O5 — Evaluation	Recorded temporal, spatial, model and API contract evidence, including held-out and limited Sri Lankan transfer comparisons.

Table 9.1 Objective-to-evidence summary for the individual component.

9.3 Individual research and product contributions

The author's research responsibility covers outbreak record harmonisation for geospatial use, relative spatial ranking, temporal and geodesic processing, vector/reach interpretation and evidence-led model reporting. The individual product responsibility covers the geospatial service boundary and its three-page veterinarian mapping experience. Team-provided diagnosis, farm-management information, user accounts and general ADRS infrastructure are integrated dependencies and are not claimed as wholly individual inventions.

9.4 Future directions

The immediate scientific priority is a sufficiently large and reliable Sri Lanka dataset with confirmed event locations and the actual times at which observations became available. A geographically representative cohort would permit disease-specific calibration, in-country holdouts and a prospective test of whether high-ranked cells anticipate real follow-on events. Additional work should validate direction bearings, evaluate historical rate transfer and define uncertainty across administrative districts. Once pre-origin weather, host density and movement factors are available across the complete evaluation cohort, an environment-aware candidate may be compared fairly with C0.

For the application, the fourteen-day map sequence should ultimately be linked to a validated day-specific model or retained as clearly illustrative scenario exploration. District ownership must be resolved from the logged-in veterinarian for every user, rather than a fixed demonstration district. Future versions can strengthen farm-level risk context, verified case lineage, saved investigation worklists, structured veterinary recommendations, and field assessment of interpretation errors and response time. Such steps would advance the component from an interpretable research-backed visualization towards an evaluated operational support tool.

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[12] R26-SE-013 geospatial component, “Individual Geospatial Thesis – Verified Implementation and Research Evidence Report,” code audit, Oct. 2026, student-supplied evidence report.

[13] Sandaru G A, “ADRS Geospatial Intelligence application demonstration,” 101-second screen recording, 9 Oct. 2026. Student-supplied demonstration artefact.

GLOSSARY

Term	Meaning in this thesis
Confirmed clinical case	Operational record designated verified by the host veterinary workflow, distinct from a predicted event.
Forecast origin	Historical disease-country and t0 context from which eligible earlier sources and later targets are defined.
Historical replay	Retrospective reconstruction using historical source-time rules and availability proxies.
Relative spatial score	Distance-based ordering of geographic cells; not an infection probability.
Direction tendency	Mathematical arrow derived from the local C0 gradient rather than an observed transmission route.
Nominal reach	S0 × day visualization descriptor, not a compulsory containment boundary.
Illustrative outlook	UI-generated forward sequence designed to communicate a scenario, not validated daily prediction.
Origin-balanced percentile	Spatial rank metric summarized with attention to independent forecast origins.

APPENDICES

Appendix A: Research artefacts and code traceability

The following table maps thesis evidence to repository modules and student-supplied artefacts. Exact line numbers may change across branches. Scientific conclusions should be checked against the final submitted commit and its local evaluation manifest before examination.

The accompanying geospatial thesis evidence archive contains 26 authentic application captures, source-code snapshots, local working-tree integration files and documented research evidence. The captured branch is review/geospatial-pages-1-2-3-confirmed at commit 310d38be15c8cf307c454cde69a21bae20d87e72 with additional uncommitted integration changes recorded in the export manifest. Application figures must be interpreted against this recorded source state.

Claim / domain	Primary evidence artefact
Raw record provenance and duplicate identities	backend/components/geospatial_tracking/data_processing/{manifest,normalize,dedup,build_canonical}.py; DATA_PROVENANCE.md
Chronology, source windows and roles	services/source_selector.py; forecast_origin.py; model_fitting_exposure.py; split_embargo.py
LSD scientific C0 kernel	services/model_development/wind_scoring_7c.py; candidate_registry_7c.py
ST-DBSCAN context	services/stdbscan/{neighborhood,cluster,config}.py
Geodesic direction	services/direction/c0_cell_local_tendency_8b3.py; CHECKPOINT_8B_EVIDENCE_SUMMARY.json
S0 and nominal reach	services/model_development/rate_s0_bootstrap_9b.py; services/integration/nominal_reach_9c.py
Scientific API and transport	api/router.py; services/transport/; protocol checkpoint evidence JSONs
Operational and vet context	repositories/host_operational_adapter.py; api/operational_router_factory.py; my_area_router_factory.py
National map and My Area views	frontend/src/features/GeospatialMap/pages/{OutbreakMapPage, MyAreaPage}.jsx
Fourteen-day presentation sequence	frontend/src/features/GeospatialMap/adapters/page1ForecastVisualization.js; myAreaPresentationForecast.js

Claim / domain	Primary evidence artefact
Analysis & Trends screen	frontend/src/features/GeospatialMap/pages/AnalysisTrendsPage.jsx
Interface visual evidence	Recording 2026-10-09 220118.mp4

Table A.1 Code and artefact traceability for examination.

Appendix B: Academic integrity and assisted-writing disclosure

The author used a code-review assistant to inspect component source paths, experimental evidence files and recorded test summaries, and used an AI-assisted drafting workflow to organize and refine the academic narrative and document formatting. The work was reconciled against the supplied research proposal, topic assessment, repository audit and application demonstration; external technical references were identified separately. AI-generated text was not treated as an experimental observation. The author remains responsible for verifying every result, quotation, source attribution, code claim and submission policy requirement before signing the declaration.

Representative assisted workflows included: (i) a read-only repository audit of the geospatial backend and frontend, (ii) comparison of the project proposal, implementation evidence and demonstration video, (iii) preparation of mathematically labelled explanatory diagrams, and (iv) formatting to the individual thesis template. No interview evidence, clinical benefit, accuracy values or new numerical experiment has been invented for this manuscript. This disclosure should be revised to match the final module AI-use policy and the student's actual workflows.

Appendix C: Demonstration walkthrough and figure provenance

The supplied recording lasts approximately 101 seconds and shows the veterinarian portal, the national Geospatial Intelligence map, a Matara-focused My Area view, a fourteen-step outlook bar chart, influencing-outbreak entries and the Analysis & Trends page. Figures reproduced from that recording are documentary screenshots of the student-provided user interface. They are not patient records or evaluation images from a statistically sampled veterinary trial.

Recording point	Visible workflow	Related thesis section
Approx. 00:25	Geospatial Intelligence national map controls	Sections 4.4.1 and 6.2.2
Approx. 00:45	Matara focused map with corridor/risk overlay	Sections 4.4.3 and 6.2.3
Approx. 00:55	Fourteen-step Future Risk Outlook and influencing sources	Sections 4.4.2, 4.4.4 and 6.2.3
Approx. 01:05	My Area summary and local control layout	Section 6.2.3
Approx. 01:15	Analysis & Trends layout	Sections 4.4.5 and 6.2.4

Table C.1 Recorded demonstration evidence by workflow.

Appendix D: Recommended validation protocol for operational extension

A subsequent prospective protocol should first freeze disease-specific case-confirmation rules, timestamps and geographic inclusion criteria with the relevant veterinary authority. It should then split prospective geographic and temporal observations without allowing case records to enter features before authorization and availability. Forecast outputs should retain exact model version, date horizon, source set, confidence semantics and performance envelope. A veterinary reviewer should independently classify whether each suggested risk region and response message is clinically useful and appropriately cautious.

Quantitative evaluation could compare the calibrated D1–D7 and D8–D14 surfaces with newly collected incident locations, reporting area-weighted rank, top-area capture, coverage, lead time and explicit missingness; directional models should additionally report angular error against verified source-target trajectories. A field task study could measure time to find a confirmed case, correct interpretation of colour and arrows, district-scoping errors, and proportion of recommendations accepted after independent clinical review. No such outcomes are claimed as results of this thesis.